MATHEMATICS

Applications and interpretation

Higher Level | Set C | Paper 1

Challenge mock examination

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A graphic display calculator and a clean Mathematics: applications and interpretation formula booklet are required.

• Answer all questions. Show working and explain mathematical reasoning in context.

• All questions are compulsory. There is no optional section.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures, unless a question specifies a different accuracy.

• Keep unrounded values in subsequent calculations. State units where relevant.

• Diagrams are not necessarily drawn to scale. Use the stated data and conditions.

• The number in square brackets gives the maximum mark for a part.

Thriving Scholars original practice | 2026 challenge series

Question 1 | Question paper

Question 1 [Maximum mark: 6]

The sound level is $L = 10\log_{10}(I/I_0)$ decibels, where $I_0 = 10^{-12}$ W m^{-2}. Two incoherent machines produce levels of 72 dB and 68 dB at a fixed location. Their intensities, rather than their decibel levels, add.

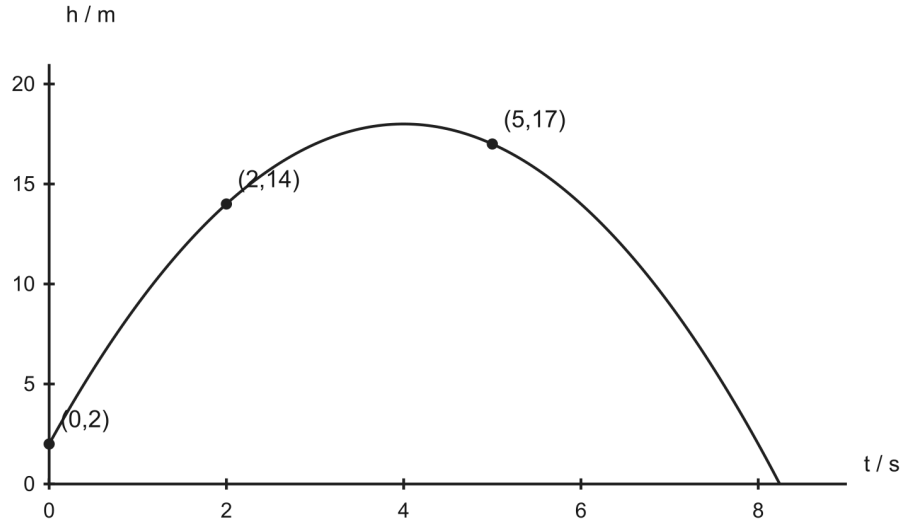
(a) Find the combined sound level when both machines operate. [3]

(b) The second machine is unchanged. Find the minimum percentage reduction in the first machine's intensity needed to make the combined level at most 70 dB. [3]

Question 2 | Question paper

Question 2 [Maximum mark: 6]

The height of a small survey drone during a programmed flight is modelled by $h(t) = at^2 + bt + c$, where t is seconds after the flight starts and h is metres above level ground. Measurements give $h(0) = 2$, $h(2) = 14$ and $h(5) = 17$. The model is used only from the start until the first contact with ground level.



(a) Find a, b, c. [3]

(b) Find the physical time domain of this model. [2]

(c) Find the maximum height predicted during this domain. [1]

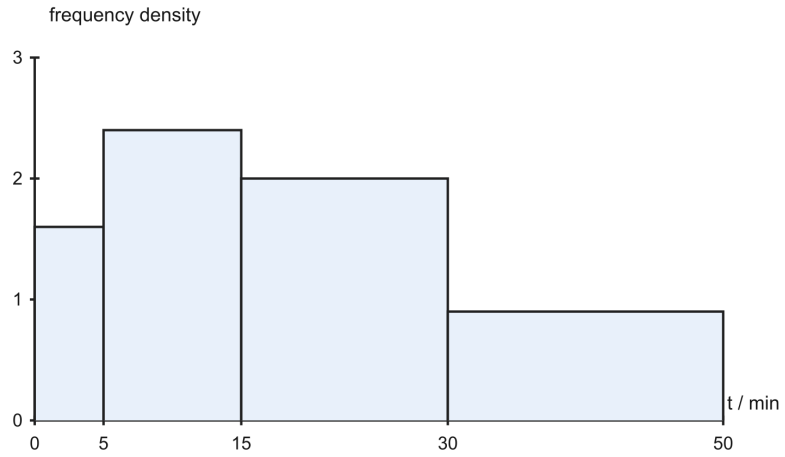
Question 3 | Question paper

Question 3 [Maximum mark: 6]

Eighty delivery delays are grouped as shown. The histogram uses frequency density on the vertical axis.

Delay t / min	Frequency
$0 \le t < 5$	8
$5 \le t < 15$	24
$15 \le t < 30$	30
$30 \le t \le 50$	18

For estimates inside a class, assume observations are uniformly distributed within that class.



(a) Calculate the height of the bar for $30 \le t \le 50$. [1]

(b) Estimate the median delay using linear interpolation. [2]

(c) Estimate the probability that a randomly selected delay exceeds 25 minutes. [2]

(d) State the modal class indicated by the histogram. [1]

Question 4 [Maximum mark: 6]

A condition affects 8% of a population. Test 1 has sensitivity 0.90 and specificity 0.85; test 2 has sensitivity 0.95 and specificity 0.98. Sensitivity is the probability of a positive result given the condition; specificity is the probability of a negative result given no condition. For each person, the two test results are independent conditional on whether that person has the condition.

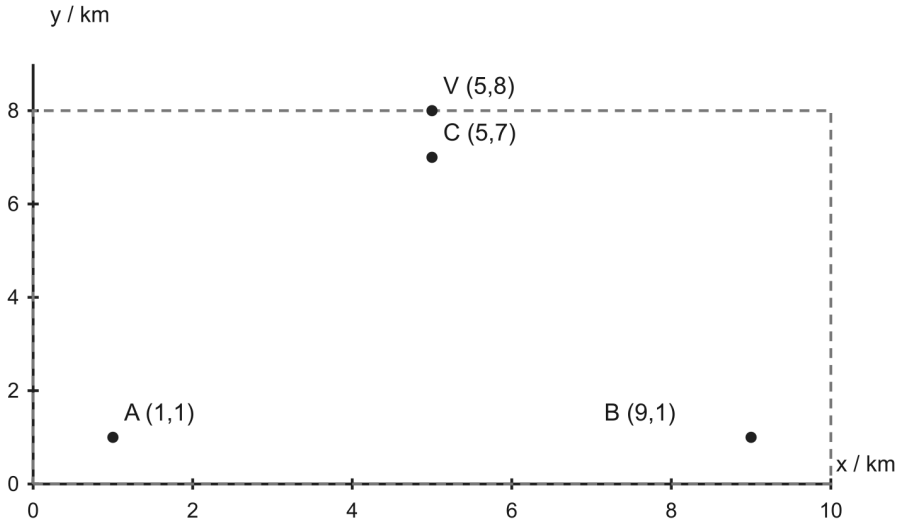
(a) Find the probability that a randomly selected person tests positive on both tests. [2]

(b) Find the probability that a person has the condition given two positive results. [2]

(c) Determine whether the two positive-test events are independent in the whole population. Support your answer numerically. [2]

Question 5 [Maximum mark: 6]

Three existing monitoring stations are at $A(1,1)$, $B(9,1)$ and $C(5,7)$ km. A rectangular reserve has $0 \leq x \leq 10$ and $0 \leq y \leq 8$. A new station will be placed either at the Voronoi vertex U equidistant from A,B,C, or at $V(5,8)$. The selected candidate should maximize its distance to the nearest existing station.



(a) Find the coordinates of U. [3]

(b) Calculate the distance from each candidate to its nearest existing station. [2]

(c) Choose between the two candidates and state the relevant comparison. [1]

Question 6 [Maximum mark: 7]

A subscriber is either a regular user R or an occasional user O each month. The regular transition matrix, with columns denoting origin states, is

$$T = \begin{pmatrix} 0.88 & 0.25 \\ 0.12 & 0.75 \end{pmatrix}$$

in the order R,O. Initially $P(R) = 0.15$. Let $p_n = P(R \text{ in month } n)$.

(a) Find a recurrence for p_n and calculate p_1. [2]

(b) Find the long-run regular-user probability. [2]

(c) Determine the first month for which the regular-user probability is at least 0.65. Justify that it is the first. [3]

Question 7 | Question paper

Question 7 [Maximum mark: 7]

Eight independent sampled pairs (x, y) are shown below. Assume they are a random sample from a bivariate normal population.

x	1	2	3	4	5	6	7	8
y	10	13	12	18	17	20	21	24

Test for positive population correlation at the 1% significance level.

(a) Calculate the Pearson correlation coefficient. [2]

(b) State the hypotheses and find the one-tailed p-value. [3]

(c) State the statistical conclusion and explain why it is not a statement that x causes y. [2]

Question 8 | Question paper

Question 8 [Maximum mark: 7]

A cumulative nutrient release R (mg) is measured t hours after treatment.

t	1	2	3	4	5	6
R	6.2	10.0	12.6	13.7	14.8	15.0

Two proposed models are $R_A = 16(1 - e^{-0.5t})$ and $R_B = -0.3t^2 + 3.5t + 1$. Define each residual as observed minus predicted.

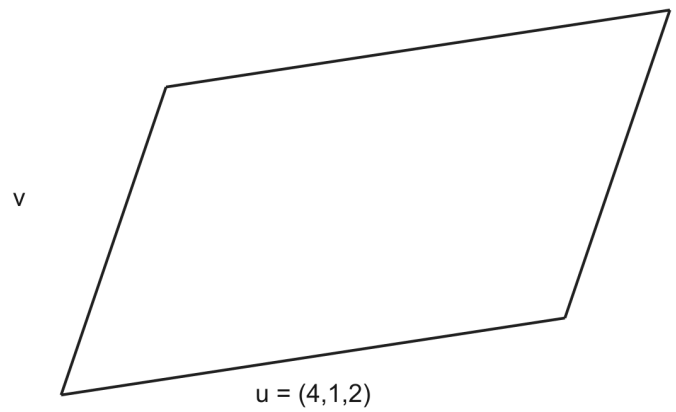
(a) Find the residuals for model A and its sum of squared residuals. [3]

(b) Find the sum of squared residuals for model B. Select the model with the better fit by this criterion. [2]

(c) Compare the physical plausibility of the models for very large t, and give a reason for caution about using either model beyond the observed times. [2]

Question 9 [Maximum mark: 7]

A parallelogram solar panel has adjacent edge vectors $\mathbf{u} = (4, 1, 2)$ and $\mathbf{v} = (1, 3, -1)$ metres. Sunlight travels in direction $\mathbf{d} = (0, -1, -2)$. The effective area facing the rays is $A_{\text{eff}} = A|\hat{\mathbf{n}} \cdot \hat{\mathbf{d}}|$, where A is panel area and hats denote unit vectors.



Schematic parallelogram; edges are 3D vectors

(a) Find a unit vector perpendicular to the panel. [3]

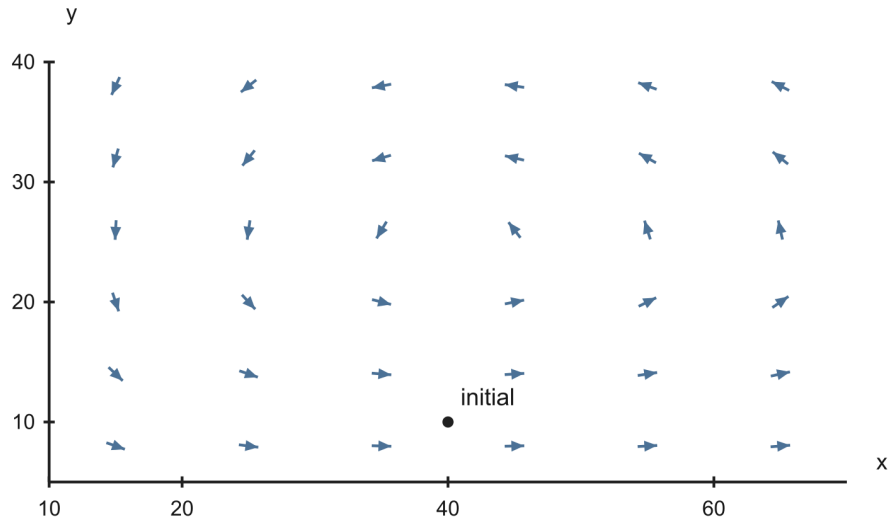
(b) Find the actual panel area. [2]

(c) Calculate the effective area facing the rays. [2]

Question 10 | Question paper

Question 10 [Maximum mark: 7]

A simplified prey-predator model is $dx/dt = 0.5x - 0.02xy$, $dy/dt = -0.4y + 0.01xy$. Populations x, y are measured in hundreds and time in months. Initially $(x, y) = (40, 10)$. The direction field is schematic; use the equations for calculations.



(a) Find the initial rate of change of each population and describe the initial direction in the phase plane. [2]

(b) Use simultaneous Euler updates of width 0.5 months to estimate both populations at $t = 1$. Show the intermediate state. [3]

(c) Find the equilibrium with both populations strictly positive. [2]

Question 11 | Question paper

Question 11 [Maximum mark: 7]

Three sectors have gross production vector $\mathbf{x}$ and final external demand $\mathbf{d}$, both in millions of dollars. Internal demand is $C\mathbf{x}$, where

$$C = \begin{pmatrix} 0.1 & 0.2 & 0.1 \\ 0.3 & 0.1 & 0.1 \\ 0.1 & 0.2 & 0.2 \end{pmatrix}$$

. Total production meets internal and external demand, so $\mathbf{x} = C\mathbf{x} + \mathbf{d}$. Initially $\mathbf{d} = (100, 80, 60)^T$.

(a) Write a matrix equation whose solution gives $\mathbf{x}$, with the coefficient matrix shown explicitly. [2]

(b) Use technology to find the required gross production vector. [3]

(c) Only external demand for sector 3 rises by 10%. Find the increase in gross production required in all three sectors. [2]

Question 12 | Question paper

Question 12 [Maximum mark: 7]

A shipment contains 80% standard cells and 20% degraded cells. Their capacities X, in arbitrary units, follow $N(100, 4^2)$ and $N(90, 6^2)$ respectively. A screening rule flags a cell if $X < c$.

(a) For $c = 94$, find the probability that a randomly selected cell is flagged. [2]

(b) Given that a cell is flagged at $c = 94$, find the probability that it is degraded. [3]

(c) Choose c so that 1% of standard cells are flagged. For this new cutoff, find the probability that a degraded cell is not flagged. [2]

Question 13 | Question paper

Question 13 [Maximum mark: 7]

Two independently and randomly sampled coatings are tested for drying time in minutes. Assume both populations are normal with equal, unknown variances.

Coating	Times
A	12.4, 11.8, 13.1, 12.7, 12.0, 13.5
B	14.1, 13.8, 15.0, 14.4, 13.5, 14.8

Test whether coating A has the lower population mean drying time at the 5% level.

(a) Find the two sample means and the pooled sample standard deviation. [3]

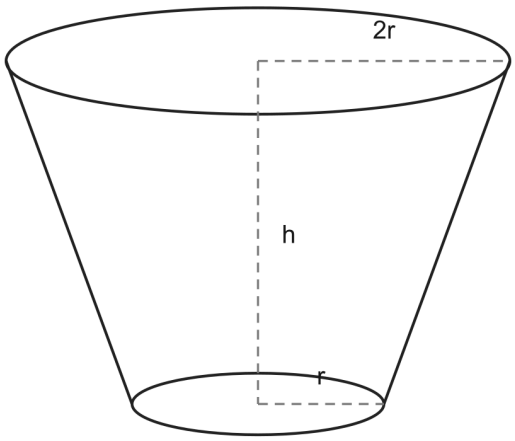
(b) State the hypotheses and calculate the test statistic and p-value. [3]

(c) State a conclusion in context. [1]

Question 14 | Question paper

Question 14 [Maximum mark: 8]

A closed storage vessel is a conical frustum with bottom radius r, top radius $2r$ and height h, all in metres. Its volume is 6 m^3. For a frustum, $V = \frac{\pi h}{3}(R^2 + Rr + r^2)$ and curved area $= \pi(R + r)\sqrt{h^2 + (R - r)^2}$. The permitted range is $0.4 \leq r \leq 1.4$.



Closed frustum; not to scale

(a) Express h in terms of r. [2]

(b) Find an expression in r for the total material area, including both circular ends. [2]

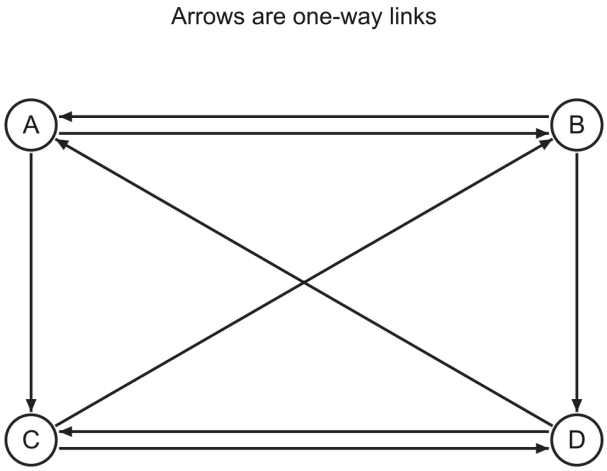
(c) Use technology to find the radius and height giving minimum total material area, and state that area. [3]

(d) Give evidence that this is the global minimum on the allowed interval. [1]

Question 15 | Question paper

Question 15 [Maximum mark: 8]

A robot moves on the directed graph shown. At each vertex it chooses uniformly between the two outgoing links. Its choices at successive steps depend only on its current vertex. Use vertex order A,B,C,D and the adjacency convention $M_{ij} = 1$ when there is an arrow from i to j.



(a) Write down the adjacency matrix M. [2]

(b) Find the number of four-link walks from A to D, identifying the relevant matrix entry. [3]

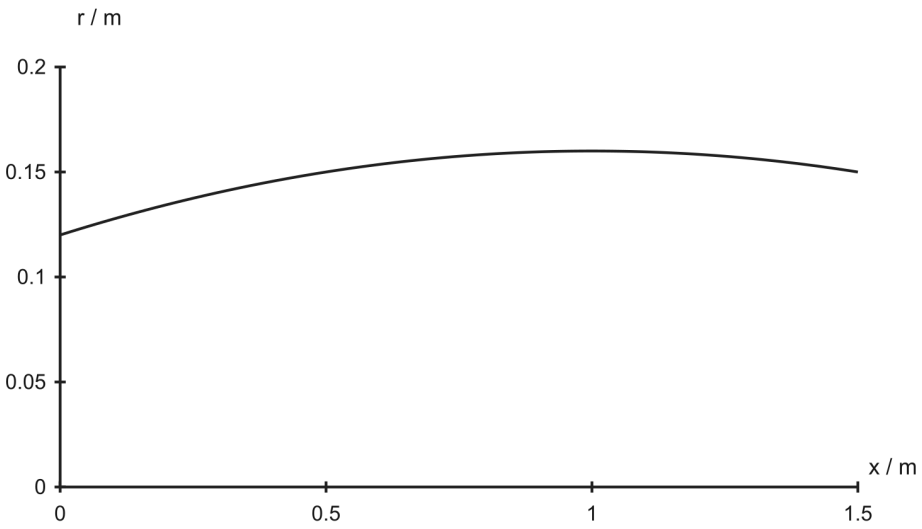
(c) Find the probability that the robot is at D after four moves, if it starts at A. [2]

(d) Explain why counting walks is appropriate even if a route revisits a vertex. [1]

Question 16 | Question paper

Question 16 [Maximum mark: 8]

A vertical vessel of height 1.5 m has circular cross-sections of radius $r(x) = 0.12 + 0.08x - 0.04x^2$ metres at height x metres above its base, for $0 \leq x \leq 1.5$. The diagram graphs radius against height; it is not a side-view orientation of the vessel. Ignore wall thickness.



(a) Find the maximum radius and the height where it occurs. [2]

(b) Calculate the vessel's capacity in litres. [3]

(c) Find the water depth when the upright vessel is half full. [3]

End of question paper

MATHEMATICS

Applications and interpretation

Higher Level | Set C | Paper 1

Worked solution book

Complete this paper before consulting the worked solutions. Each part has method (M), accuracy (A) and reasoning (R) points. The number following a code is its mark value. Award only the points supported by the candidate's retained working.

Accept mathematically equivalent valid methods. Numerical answers may use technology, but should be supported by equations, test definitions, bounds, labelled graphs or other appropriate mathematical working. Retain unrounded intermediate values.

Use the portal to enter a whole-number self-assessed mark for each part. These are independent practice papers; the self-assessed report does not calculate an official IB grade.

Question 1 | Worked solutions

Question 1 [Maximum mark: 6]

(a) [3 marks]

M1 Convert to $I_1 = I_0 10^{7.2}$ and $I_2 = I_0 10^{6.8}$.

M1 Use $L = 10 \log_{10}[(I_1 + I_2)/I_0]$.

A1 The combined level is 73.455405 dB.

(b) [3 marks]

M1 A 70 dB limit corresponds to total intensity 10^{-5} W m^{-2}.

M1 The retained fraction k satisfies $kI_1 + I_2 = 10^{-5}$.

A1 $k = 0.23285017$, so reduce the first intensity by at least 76.714983%.

Question 2 | Worked solutions

Question 2 [Maximum mark: 6]

(a) [3 marks]

M1 Set up $c = 2$, $4a + 2b + c = 14$, $25a + 5b + c = 17$.

M1 Solve the simultaneous equations.

A1 $a = -1, b = 8, c = 2$.

(b) [2 marks]

M1 Solve $-t^2 + 8t + 2 = 0$ and retain the positive root.

A1 $0 \leq t \leq 8.2426407$ seconds.

(c) [1 marks]

A1 The vertex is at $t = 4$, giving maximum height 18 m.

Question 3 [Maximum mark: 6]

(a) [1 marks]

A1 The density is $18/20 = 0.9$ observations per minute.

(b) [2 marks]

M1 The median is the 40th cumulative observation, in the 15-to-30 class after 32 observations.

A1 $15 + [(40 - 32)/30](15) = 19$ minutes.

(c) [2 marks]

M1 Use one third of the 30 observations in 15-to-30, plus all 18 in the last class.

A1 The estimated probability is 0.35.

(d) [1 marks]

A1 The modal class is $5 \leq t < 15$, which has the greatest frequency density.

Question 4 [Maximum mark: 6]

(a) [2 marks]

M1 Use $0.08(0.90)(0.95) + 0.92(0.15)(0.02)$.

A1 The probability is 0.07116.

(b) [2 marks]

M1 Divide $0.08(0.90)(0.95)$ by the probability in part (a).

A1 The posterior probability is 0.96121417.

(c) [2 marks]

M1 $P(+_1) = 0.21$, $P(+_2) = 0.0944$; their product is 0.019824.

R1 This differs from $P(+_1 \cap +_2) = 0.07116$, so the positive events are not unconditionally independent. Conditional independence does not imply unconditional independence.

Question 5 [Maximum mark: 6]

(a) [3 marks]

M1 The perpendicular bisector of AB is $x = 5$.

M1 Equating distances to A and C gives $8x + 12y = 72$.

A1 $U = (5, 8/3)$.

(b) [2 marks]

A1 At U the nearest distance is $13/3$ km.

A1 At V the nearest station is C, at distance 1 km.

(c) [1 marks]

R1 Choose U, since $13/3 > 1$. This compares only the two permitted candidates, not every point in the reserve.

Question 6 [Maximum mark: 7]

(a) [2 marks]

M1 $p_{n+1} = 0.88p_n + 0.25(1 - p_n) = 0.63p_n + 0.25.$

A1 $p_1 = 0.3445.$

(b) [2 marks]

M1 Solve $p = 0.63p + 0.25.$

A1 $p = 0.67567568.$

(c) [3 marks]

M1 Iterate the recurrence or use $p_n = p_* + (0.15 - p_*)(0.63)^n.$

A1 $p_6 = 0.64280859 < 0.65$ and $p_7 = 0.65496941 \geq 0.65.$

R1 The first month is 7; the sequence is increasing toward $p_* > 0.65.$

Question 7 [Maximum mark: 7]

(a) [2 marks]

M1 Use all eight raw pairs in a Pearson calculation.

A1 $r = 0.96737271$.

(b) [3 marks]

A1 $H_0 : \rho = 0, H_1 : \rho > 0$.

M1 Use the correlation test with $n = 8$ (equivalently $t = r\sqrt{6/(1-r^2)}$, $df = 6$).

A1 $t = 9.3526703$, $p = 4.2360842e-05$.

(c) [2 marks]

R1 $p = 4.2360842e-05 < 0.01$, so reject zero correlation in favour of positive correlation.

R1 Even significant correlation does not identify a causal mechanism or remove possible common causes.

Question 8 [Maximum mark: 7]

(a) [3 marks]

M1 Evaluate model A at all six times and subtract from the observations.

A1 The residuals are $[-0.095509, -0.113929, 0.170083, -0.134635, 0.11336, -0.203407]$ mg.

A1 $SS_A = 0.1233815\ \text{mg}^2$.

(b) [2 marks]

A1 $SS_B = 69.81\ \text{mg}^2$.

R1 Choose model A because its residual sum of squares is smaller.

(c) [2 marks]

R1 A approaches 16 mg, whereas B eventually decreases and becomes negative, which is implausible for cumulative release.

R1 The data only cover 1 to 6 hours; the long-term mechanism or environment may differ, so the better observed fit does not validate distant extrapolation.

Question 9 | Worked solutions

Question 9 [Maximum mark: 7]

(a) [3 marks]

M1 Calculate $\mathbf{u} \times \mathbf{v} = (-7, 6, 11)$.

M1 Its magnitude is $\sqrt{206}$.

A1 One unit normal is

$$\begin{pmatrix} -0.48771311 \\ 0.41803981 \\ 0.76640632 \end{pmatrix}$$

; its negative is also valid.

(b) [2 marks]

M1 $A = |\mathbf{u} \times \mathbf{v}|$.

A1 $A = 14.3527\ \text{m}^2$.

(c) [2 marks]

M1 $A_{\text{eff}} = |(-7, 6, 11) \cdot (0, -1, -2)|/\sqrt{5}$.

A1 $A_{\text{eff}} = 12.521981\ \text{m}^2$.

Question 10 [Maximum mark: 7]

(a) [2 marks]

A1 $x\prime(0) = 12, y\prime(0) = 0.$

R1 The trajectory initially moves horizontally toward increasing x.

(b) [3 marks]

M1 At each step evaluate both derivatives at the same old pair.

A1 The first state is $[46.0, 10.0]$.

A1 The $t = 1$ state is $[52.9, 10.3]$ hundreds.

(c) [2 marks]

M1 For positive populations solve $0.5 - 0.02y = 0$ and $-0.4 + 0.01x = 0$.

A1 The coexistence equilibrium is $(40, 25)$.

Question 11 | Worked solutions

Question 11 [Maximum mark: 7]

(a) [2 marks]

M1 $(I - C)\mathbf{x} = \mathbf{d}$.

A1 The coefficient matrix is

$$\begin{pmatrix} 0.9 & -0.2 & -0.1 \\ -0.3 & 0.9 & -0.1 \\ -0.1 & -0.2 & 0.8 \end{pmatrix}$$

(b) [3 marks]

M1 Use $\mathbf{x} = (I - C)^{-1}\mathbf{d}$.

A2

$$\mathbf{x} = \begin{pmatrix} 161.06195 \\ 157.52212 \\ 134.51327 \end{pmatrix}$$

million dollars.

(c) [2 marks]

M1 $\Delta\mathbf{x} = (I - C)^{-1}(0, 0, 6)^T$.

A1 The increase is

$$\begin{pmatrix} 1.1681416 \\ 1.2743363 \\ 7.9646018 \end{pmatrix}$$

million dollars.

Question 12 [Maximum mark: 7]

(a) [2 marks]

M1 Use $0.8P(X < 94 \mid \text{standard}) + 0.2P(X < 94 \mid \text{degraded})$.

A1 The probability is 0.20294725.

(b) [3 marks]

M1 The degraded-and-flagged probability is $0.2\Phi((94 - 90)/6) = 0.14950149$.

M1 Divide by the total flagging probability.

A1 The posterior probability is 0.73665196.

(c) [2 marks]

A1 $c = 100 + 4\Phi^{-1}(0.01) = 90.694609$.

A1 $P(X \geq c \mid \text{degraded}) = 0.45391817$.

Question 13 | Worked solutions

Question 13 [Maximum mark: 7]

(a) [3 marks]

A1 $\bar{x}_A = 12.583333$, $\bar{x}_B = 14.266667$ minutes.

M1 $s_p = \sqrt{[5s_A^2 + 5s_B^2]/10}$.

A1 $s_p = 0.61495257$ minutes.

(b) [3 marks]

A1 $H_0 : \mu_A = \mu_B$, $H_1 : \mu_A < \mu_B$.

M1 $t = (\bar{x}_A - \bar{x}_B)/(s_p\sqrt{1/6 + 1/6})$, with $df = 10$.

A1 $t = -4.7412093$, one-tailed $p = 0.00039533172$.

(c) [1 marks]

R1 $p = 0.00039533172 < 0.05$, so there is evidence that coating A has a lower population mean drying time.

Question 14 | Worked solutions

Question 14 [Maximum mark: 8]

(a) [2 marks]

M1 $6 = (7\pi/3)hr^2$.

A1 $h = 18/(7\pi r^2)$.

(b) [2 marks]

M1 The end areas total $5\pi r^2$.

A1 $S(r) = 5\pi r^2 + 3\pi r\sqrt{r^2 + [18/(7\pi r^2)]^2}$.

(c) [3 marks]

M1 Minimize $S(r)$ numerically over $0.4 \le r \le 1.4$.

A1 $r = 0.59197481$ m, $h = 2.3357058$ m.

A1 The minimum area is $18.948072\ \text{m}^2$.

(d) [1 marks]

R1 The derivative changes from negative to positive at its only zero in the interval; endpoint areas are 21.857853 and $50.064485\ \text{m}^2$, both larger. A suitable GDC derivative graph is acceptable.

Question 15 | Worked solutions

Question 15 [Maximum mark: 8]

(a) [2 marks]

A2 $$M = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$. Award one mark for two correctly specified rows and both marks for all four.

(b) [3 marks]

M1 Calculate M^4.

M1 Use the entry in row A, column D.

A1 The number is 4.

(c) [2 marks]

M1 Every four-move walk has probability $(1/2)^4$.

A1 The probability is $4/16 = 0.25$.

(d) [1 marks]

R1 Walks permit repeated vertices and links; a robot can revisit them, unlike a simple path.

Question 16 | Worked solutions

Question 16 [Maximum mark: 8]

(a) [2 marks]

M1 $r\prime(x) = 0.08 - 0.08x = 0$ gives $x = 1$.

A1 The maximum radius is 0.16 m, greater than both endpoint radii.

(b) [3 marks]

M1 $V = \pi \int_0^{1.5} (0.12 + 0.08x - 0.04x^2)^2 \, dx$.

A1 $V = 0.10659424 \text{ m}^3$.

A1 The capacity is 106.59424 litres.

(c) [3 marks]

M1 Solve $\pi \int_0^h r(x)^2 \, dx = V/2$ over $0 < h < 1.5$.

M1 Use a numerical equation solver with the full unrounded capacity.

A1 The depth is $h = 0.81582338$ m.

End of worked solution book